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Research

Solid-state methods for ceramic metallization: Unlocking structural complexity, quantum effects, and multifunctionality



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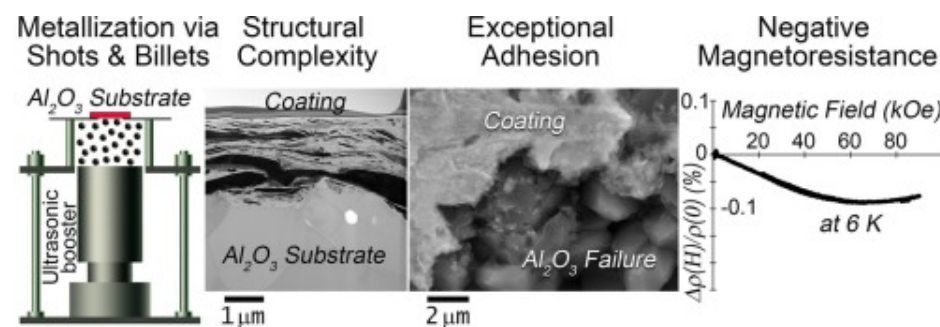
Highlights

- Solid-state metallization of ceramics via milling shots and billets.
- Formation of structurally complex, integrated metal-ceramic systems.
- Coatings impart conductivity and magnetism to insulating ceramics.
- Exceptional adhesion ensures mechanical integrity under deformation.
- Weak localization and negative magnetoresistance reveal quantum transport phenomena.

Abstract

The metallization of ceramics is essential for advanced technological applications but remains challenging due to poor metal-ceramic adhesion and structural incompatibility. A solid-state mechanical alloying approach using ultrasonically assisted shot impact processing, which can also utilize metallic cubic billets as a coating material source, is introduced. This approach is demonstrated on W-Al- and W-Ni-based coatings fabricated on alumina substrates. The method enables the formation of a structurally complex system, integrating amorphous regions, nanocrystalline grains, and localized non-equilibrium solid solutions into a uniform structure. The as-fabricated coatings impart electrical conductivity and magnetic properties to the initially insulating ceramic substrate, while their exceptional adhesion ensures mechanical integrity, allowing the coating to co-deform with the substrate without delamination prior to ceramic fracture. Additionally, quantum transport phenomena, including weak localization and negative magnetoresistance at 6K, are observed, revealing a direct correlation between structural complexity and electronic behavior. The emergence of quantum phenomena in mechanically integrated multicomponent metallic systems provides insight into quantum behavior in severely deformed materials and opens new possibilities for designing advanced functional materials.

Graphical abstract



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Discussion

Conclusions

CRediT authorship contribution statement

Declaration of competing interest

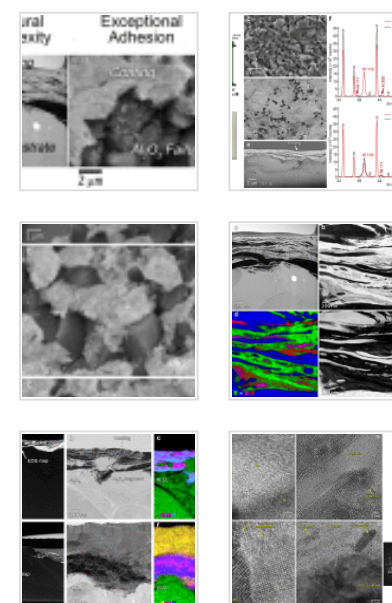
Acknowledgements

Data availability

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Keywords

Ceramic metallization; Solid-state processing; Multicomponent coatings; Microstructure; Adhesion; Quantum transport; Multifunctionality

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Introduction

Modern materials are increasingly reaching their performance limits, creating a critical need for hybrid materials with tailored properties [[1], [2], [3], [4], [5], [6], [7]]. One of the most promising strategies in this regard focuses on composite materials inspired by natural systems, which often exhibit hierarchical and multifunctional structures [[8], [9], [10], [11], [12], [13]]. These bio-inspired materials exemplify the potential of hybrid designs to achieve synergies between components, unlocking unique properties that surpass those of their individual constituents. Multifunctionality in these systems is achieved through the integration of diverse materials into unified structures, enabling them to simultaneously fulfill mechanical, thermal, electrical, and magnetic demands as well as exhibit radiation tolerance [6,7,[14], [15], [16], [17], [18], [19], [20], [21]]. A critical challenge in advancing this field lies in developing techniques to combine materials with fundamentally different physical and chemical properties, such as ceramics and metals. Among these techniques, the metallization of ceramics—a process that applies a thin metal layer onto ceramic substrates to enhance their functionality—has emerged as particularly significant [[22], [23], [24], [25], [26]]. This method facilitates the development of multifunctional systems in which ceramic substrates provide thermal stability and mechanical strength, while metallic layers add conductivity, magnetic properties, and serve as platforms for further functionalization.

The metallization of ceramics is critically important for applications in electronics, sensors, energy, aerospace, and telecommunications [22]. However, metallization of ceramics presents significant challenges, with adhesion being a primary concern [[26], [27], [28], [29], [30], [31], [32], [33], [34], [35]]. Ceramic surfaces are typically non-conductive and inert, posing difficulties for strong metal bonding. Widely used methods such as physical vapor deposition (PVD) and chemical vapor deposition (CVD) provide partial solutions but often struggle to achieve a combination of strong adhesion and efficient processing. Material compatibility adds another layer of complexity: the selected metal must align with the specific ceramic to avoid undesirable reactions or weak adhesion interfaces. These compatibility issues become particularly pronounced when fabricating complex multicomponent metallic coatings on ceramic substrates, where conventional methods often face significant limitations. For instance, materials such as tungsten and aluminum exemplify these challenges. Due to their distinct properties, co-deposition of these metals on ceramic surfaces using PVD or CVD methods remains highly problematic, underscoring the need for alternative metallization strategies. Conventional techniques also involve high energy consumption, production costs, and the use of hazardous precursor chemicals, creating environmental concerns. Solid-state methods have emerged as a promising solution [[32], [33], [34], [35]], bypassing many of these limitations while enabling the development of new hybrid multifunctional materials and advanced metallization strategies.

To address the challenges of conventional metallization techniques, we developed an innovative approach using ultrasonically assisted shot impact equipment, where the coating material is directly sourced from the impacting bodies themselves. This method builds upon our previous solid-state vibratory milling process, which utilized a combination of milling balls to fabricate composite coatings with high adhesion on metallic substrates [[36], [37], [38]]. In our novel modification, mechanical vibrations were replaced with ultrasonic oscillations as the source of impact energy, making the process more adaptable for ceramic substrates and enabling the treatment of larger areas. Additionally, we substituted conventional spherical milling balls, which are complex and

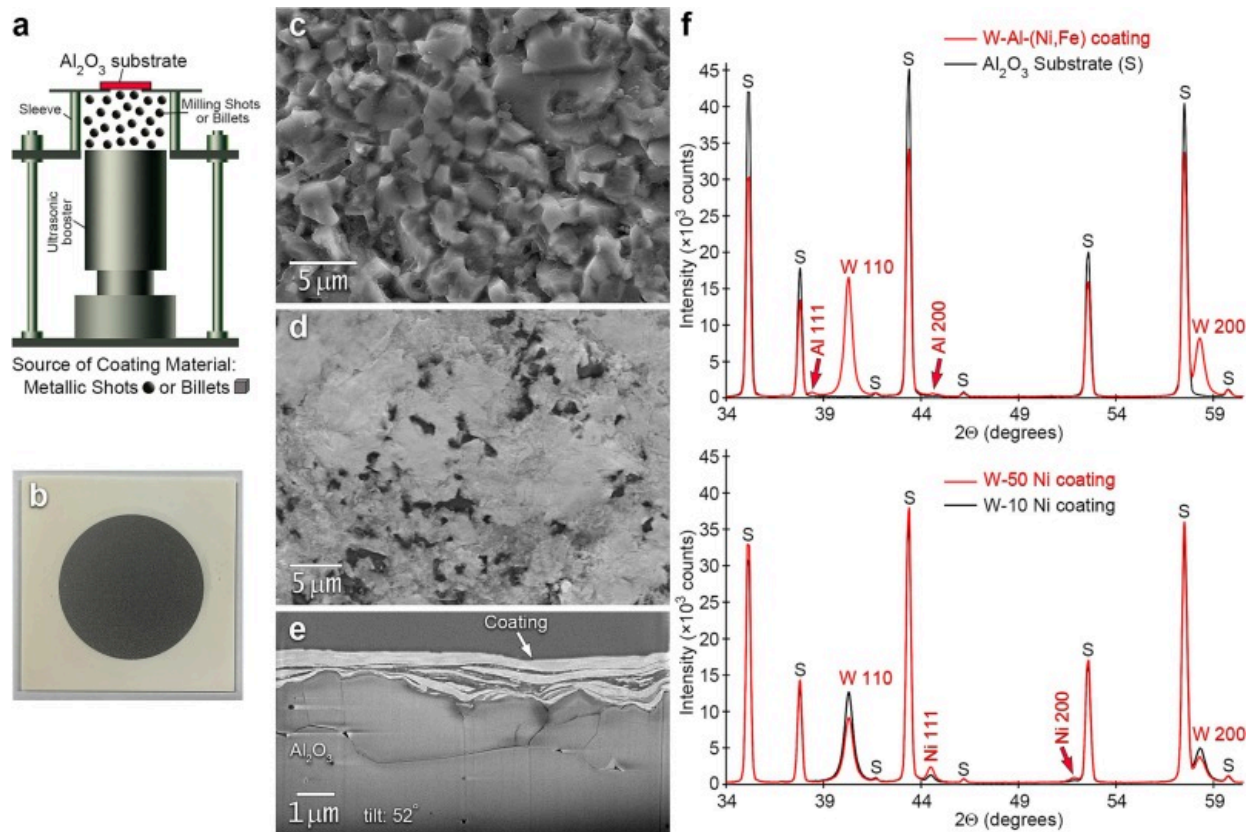
costly to produce, with cubes cut from billets. This substitution not only simplifies the process but also expands the range of milling media that can be employed as coating material sources. Operating at ambient temperature and without requiring special atmospheric conditions, the combination of ultrasonic oscillations and flexible material selection offers enhanced control and adaptability, paving the way for the creation of advanced hybrid systems with tailored properties.

In this work, we focus on the fabrication of composite coatings on alumina substrates using shot impact equipment, with tungsten (W) as the primary coating material combined with aluminum (Al) or nickel (Ni). The selection of materials was guided by both fundamental and applied considerations. Tungsten, known for its high thermal conductivity, melting point, and exceptional mechanical strength, was paired with either highly conductive Al or magnetic Ni to produce composite structures that are difficult to fabricate using conventional methods. These combinations hold particular promise for the development of innovative materials exhibiting quantum effects, as discussed in this work. From a practical standpoint, W-based structures with high thermal conductivity on insulating ceramics are especially appealing for electronic devices, where efficient heat dissipation is critical to prevent overheating and ensure reliable performance [25]. The goal of this work is to demonstrate the potential of ultrasonically assisted shot impact equipment in fabricating composite coatings on ceramic substrates that overcome traditional limitations, while providing new opportunities for advanced material design.

Experimental details

The ultrasonically assisted shot impact equipment was developed as an alternative method for coating fabrication, particularly for cold welding pre-deposited powder to a substrate [39,40]. The current process variant is schematically presented in Fig. 1a. The system utilizes a titanium booster with a top diameter of 70mm, connected to a piezoceramic transducer (Hielscher UIP1000hd, Germany). This setup generates vertical ultrasonic vibrations with an amplitude of up to 80 μ m at a frequency of 20kHz. These vibrations accelerate the shots (or balls) placed above the booster (Fig. 1a). The

accelerated shots collide with a cylindrical stainless steel sleeve positioned above the booster. This sleeve confines the motion of the shots within the treatment zone and directs them towards the surface of the workpiece, which is fixed above the sleeve.



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Fig. 1. Fabrication process and structural characterization of the coatings: (a) schematic illustration of the fabrication process; (b) surface appearance of the metallized alumina substrate; (c) microstructure of the as-received ceramic substrate; (d) SEM micrograph showing a representative coating morphology; (e) cross-sectional FIB-prepared SEM image of the W-Al-(Ni,Fe) coating; (f) XRD patterns of the as-fabricated coatings.

In our earlier work, powder suspensions were applied to the substrate surface, dried, and processed in multiple cycles, introducing additional operations and complexities that made the approach less practical for industrial applications. Specifically, preparing multicomponent suspensions with consistent ratios was problematic due to differences in powder density and sedimentation behavior. Achieving a uniform distribution of the suspension over the substrate surface was particularly demanding due to the manual nature of the pre-deposition process, which often resulted in uneven layer thickness and poor adhesion of the powder during drying and handling. If the thickness of the precoated layer exceeded a certain threshold, the powder particles could not cold weld to the substrate during impact processing. Instead, loose powder detached and accumulated within the working space, diminishing the milling intensity and impact energy. To ensure effective particle welding, only thin precoated layers could be applied, requiring multiple precoating and processing cycles to fabricate a thick and continuous film. This significantly reduced process efficiency and posed challenges for industrial automation. Moreover, the handling of fine metallic powders introduced additional risks, as many are prone to oxidation or are potentially hazardous and explosive in certain processing environments. In contrast, the current approach eliminates the need for powders or multiple processing cycles by utilizing the impacting bodies themselves as the material source ([Fig. 1a](#)). This concept simplifies the process and enhances its potential for industrial applications.

Alumina specimens (Al_2O_3) with dimensions of $50 \times 50 \text{ mm}^2$ and a thickness of 3 mm were used as substrates. All consumable materials used in this study were procured through As One Corporation (Japan), a supplier of laboratory-grade materials. The substrates were placed on a stainless steel holder with a 30 mm diameter hole, positioned on top of the cylindrical sleeve. For the fabrication of W-Al composite coatings, we used a combination of 46 g of W shots (184 pieces) and 10 g of Al shots (250 pieces), each 3 mm in diameter. The W shots used in this study correspond to W-Ni-Cu metal matrix composites with a nominal composition of W-2.5Cu-2.5Ni-1Fe wt%, where iron is considered a trace impurity, and the binder phase, composed of a combination of Ni and Cu, accounts for

approximately 8–10vol% of the material. The Al shots were made of aluminum alloy 5052 (Al-2.2Mg-0.4Fe-0.2Cr wt%), with magnesium acting as the primary alloying element. To further refine and simplify the method, a separate set of experiments was conducted for the deposition of W-Ni coatings, where Ni cubes ($3\times3\times3\text{ mm}^3$), cut from long Ni billets (99.95% pure), were used instead of Al shots (Fig. 1a). The following combinations were employed: 46g of W shots (184 pieces) with 11g of Ni cubic billets (50 pieces), and 28g of W shots (112 pieces) with 28g of Ni cubic billets (127 pieces). Based on experimental observations, a total milling body mass of approximately 50–60 g provided the highest impact intensity under the specific working space configuration used in this study. Therefore, a total mass around 55 g was selected as the standard condition, considering minor variations in individual shot and billet weights. The impact treatment of alumina substrates was carried out at room temperature without any special atmospheric conditions, with a processing duration of 30 min.

The phase composition of the coatings was analyzed using X-ray diffraction (XRD) on a Rigaku SmartLab diffractometer with $\text{CuK}\alpha$ radiation. Surface roughness was measured using a SURFCOM 1400G surface texture measurement instrument (ACCRETECH, Tokyo Seimitsu Co.). The surface microstructure was examined using a JEOL JSM-7800F field-emission scanning electron microscope (SEM). To study the coating thickness and cross-sectional structure, a series of cross-section samples were prepared and analyzed using a focused ion beam (FIB) instrument (FEI Versa 3D FIB). Transmission electron microscopy (TEM) was performed on cross-sectional samples prepared with a FEI Helios NanoLab DualBeam FIB system. The sample preparation methodology for TEM is described in [41]. TEM observations were conducted using an atomic-resolution analytical electron microscope (JEOL JEM-ARM200F) equipped with a Quantax (Bruker) energy-dispersive spectroscopy (EDS) system for microanalysis. The structure of the specimens was analyzed using various TEM modes, including high-resolution TEM (HRTEM), bright-field (BF) TEM, and high-angle annular dark-field (HAADF) scanning TEM (STEM). All TEM images presented in this work were acquired from cross-sectional samples.

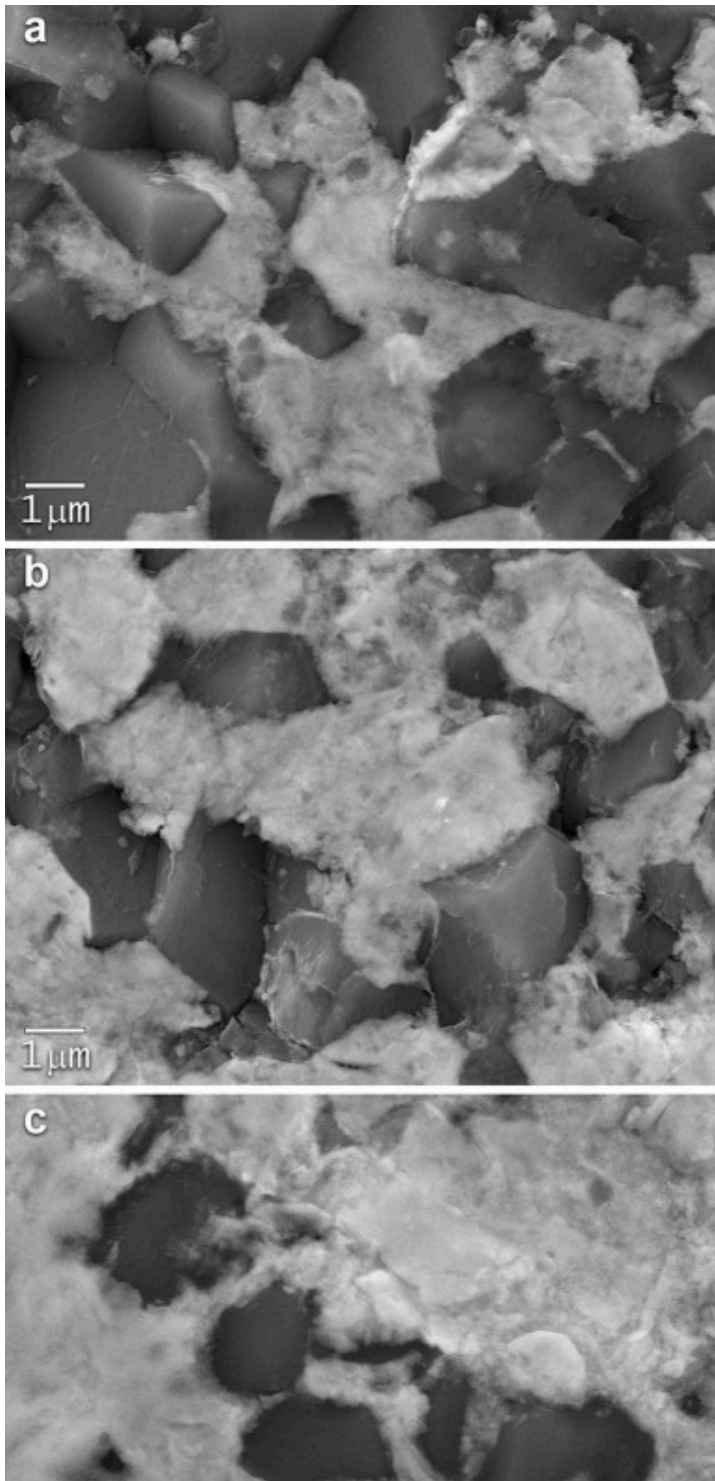
Scratch testing was conducted using an RST3 Scratch Tester (Anton Paar) equipped with a diamond-tipped indenter of 200 μ m radius. Scratches were made over a 3 mm track, starting with an initial load of 1 N, which progressively increased to 200 N at a loading rate of 398 N/min and a traverse speed of 6 mm/min. Three scratches were produced on each sample. The resulting surface morphology was analyzed using SEM, and cross-sectional views of the scratches were prepared using FIB milling perpendicular to the scratch direction. For comparative analysis, results were benchmarked against a Diamond-Like Carbon (DLC) coating (BCR-692 reference material) and an industrial TiN (Titanium nitride) coating deposited on tool steel.

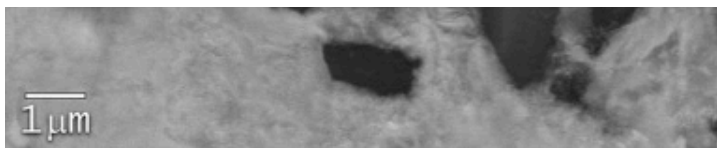
Magnetization measurements of the as-fabricated specimens were performed at room temperature using a vibrating sample magnetometer (VSM-5, Toei Industry Co., Ltd.) under an applied magnetic field of up to 16 kOe. Electrical resistivity of the coating fabricated on the ceramic substrate was determined by the conventional four-probe method using a Physical Property Measurement System (PPMS, Quantum Design Ltd.) over a temperature range of 6 to 300 K. Additionally, electrical resistivity was measured under an applied magnetic field of up to 90 kOe at 6 K to evaluate magnetoresistance.

Results

Microstructure and formation mechanisms

The experimental setup described here enables the deposition of composite coatings on ceramic substrates (Fig. 1b). To elucidate the mechanisms of coating formation, it is crucial to analyze the structural characteristics. As illustrated in Fig. 1c, the substrate consists of polycrystalline grains with irregular polygonal shapes and well-defined facets (see also Fig. 2). These grains protrude from the ceramic surface, exposing sharp edges and faceted grooves that contribute to surface roughness and heterogeneity. This distinctive morphology likely plays a critical role in the material transfer process during impact.





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Fig. 2. Sequential stages of the coating formation process illustrated by SEM images: (a) material anchoring; (b) filling of intergranular grooves and valleys; (c) spreading and leveling of surface imperfections. Images (a) and (b) were obtained from supplementary experiments using only W shots. The limited plasticity of pure tungsten restricts cold welding, enabling clearer observation of the initial anchoring stages. The observed morphologies are not associated with a specific processing time but represent the general mechanism of coating formation.

When the impacting bodies collide with the ceramic substrate, the sharp edges and irregularities of the polycrystalline grains promote localized deformation and material transfer from the softer impacting bodies, initiating material anchoring and the coating formation process (Fig. 1d). As illustrated in Fig. 2a, material fragments from the milling media initially become lodged between the ceramic grain facets, creating anchor points for subsequent layers. Such localized attachment facilitates further cold welding during subsequent impacts. Gradually, the material spreads, filling grooves and valleys between the grains (Fig. 2b), leveling surface imperfections and leaving only faint traces of the most prominent ceramic grains visible through the coating (Fig. 2c). Under continued impact, the adhered material fragments are compacted and consolidated into a unified, monolithic coating (Fig. 1d). A typical cross-sectional view (Fig. 1e) shows the coating tightly filling intergranular grooves and effectively leveling structural irregularities caused by the exposed facets of the ceramic grains on the surface. While this process smoothens the surface at the fine grain scale, the localized welding and compaction of material fragments, combined with residual indentations from the impacting bodies, result in a wavy, island-like morphology at a microscale level, increasing the overall

surface roughness (Fig. 1d and 2). For the initial ceramic substrate, the R_a value was $0.178\mu\text{m}$, while the as-fabricated coatings exhibit roughness values in the range of 0.24 to $0.37\mu\text{m}$ (Table 1). This increase remains within the acceptable range for metallic surfaces typically observed in other coating processes, such as chemical deposition or electroplating, where surface roughness can be considerably higher [22,42]. The variation in roughness among the coatings is attributed to differences in composition, mechanical properties of the components, and coating thickness (Table 1). The thickness of the coatings ranged between 600 and 1500nm , with an average value of approximately 1000nm . This variation is mainly influenced by the initial morphology of the substrate, particularly the grooves and valleys between ceramic grains, as observed in the cross-sectional images (Fig. 1e).

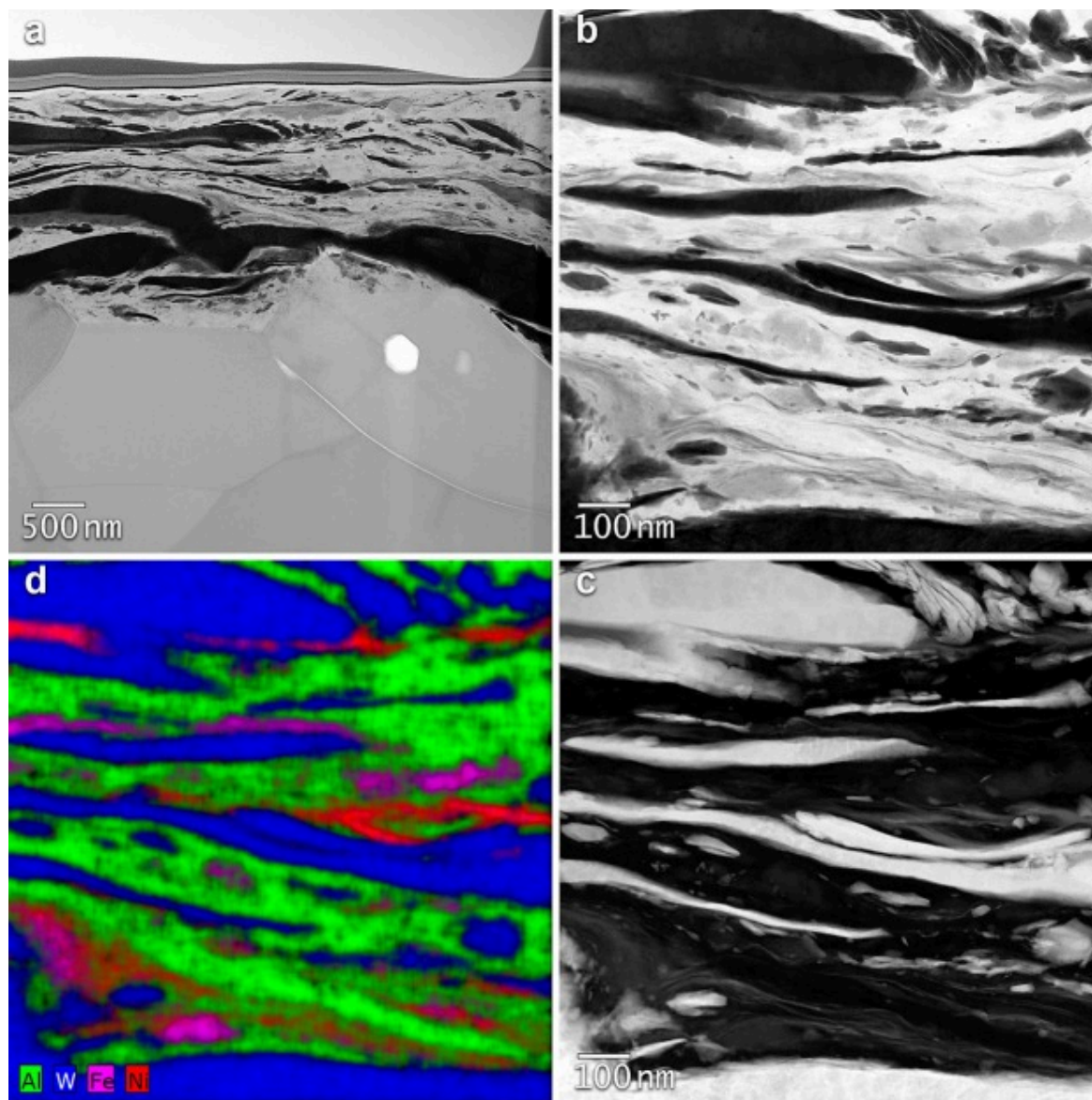
Table 1. Composition, component ratios, roughness, and thickness of the as-fabricated coatings.

	W-Al-(Ni,Fe)	W-10Ni	W-50Ni
Mass fraction	Al: 10 – 15wt%		
	W: 75 – 85wt%	W: 85 – 90wt%	W: 45 – 50wt%
	Ni: 4 – 5wt%	Ni: 5 – 10wt%	Ni: 45 – 50wt%
	Fe: 4 – 5wt%	Fe: 2 – 4wt%	Fe: 2 – 4wt%
	Cu: 2 – 4wt%	Cu: 2 – 3wt%	Cu: 1 – 2wt%
Volume fraction	Al: 40 – 50%		
	W: 35 – 45%	W: 70 – 80%	W: 30 – 40%
	Ni: 5 – 7%	Ni: 10 – 20%	Ni: 60 – 70%
	Fe: 5 – 7%	Fe: 5 – 7%	Fe: 5 – 7%
	Cu: 2 – 4%	Cu: 2 – 4%	Cu: 1 – 2%

	W-Al-(Ni,Fe)	W-10Ni	W-50Ni
Volume fraction ratios	Al (50%): W (35–40%)	W (70–80%): Ni (20%)	W (40%): Ni (60–70%)
	1.25 – 1.42	3.5 – 4	0.57 – 0.66
Number ratios of milling bodies	Al: W = 1.35	W: Ni=3.68	W: Ni=0.88
Ra, μm (Initial: 0.178$\pm$0.009)	0.377 $\pm$ 0.014	0.255 $\pm$ 0.009	0.240 $\pm$ 0.009
Thickness, nm (range/mean)	1000 – 1500/ 1220	600 – 1300/ 1020	700 – 1400/ 1100

The coatings exhibit a composite structure, indicating successful material incorporation and consolidation during the process. [Fig. 3](#) highlights the multicomponent nature of the coatings, showing the integration of various elements into a unified, complex structure. The chemical composition and volume fractions of the components are summarized in [Table 1](#). For the coating produced using W and Al shots, aluminum and tungsten occupy approximately equal volume fractions, each comprising 40–50% of the coating. [XRD analysis](#) confirms the presence of face-centered cubic (fcc) Al and body-centered cubic (bcc) W phases, highlighting the composite nature of the coatings ([Fig. 1f](#)). The remaining elements, Ni and Cu, originate from the W shots used during the process. Fe, detected in concentrations of 2–5 wt% ([Table 1](#)), primarily comes from the stainless steel sleeve and holder ([Fig. 1a](#)), a common issue in ball milling processes [[43](#)], [[44](#)], [[45](#)], [[46](#)]]. This contamination could be reduced by using sleeves and holders made of alternative materials. Despite their low concentrations, Cu and Fe are consistently present in all coatings ([Table 1](#)), but are not detected by XRD due to their low levels, similar to the Ni component in the W-Al coating ([Fig. 1f](#)). While the original [Al alloy](#) (5052) shot contains ~2.2 wt% Mg, the resulting coatings exhibited only trace amounts of magnesium

(~0.1 wt%), as revealed by EDS measurements. Notably, the presence of Fe and Ni in the W-Al coating contributes additional functional properties, as discussed below. For this reason, the as-fabricated W-Al coating is referred to as W-Al-(Ni,Fe) throughout this work.

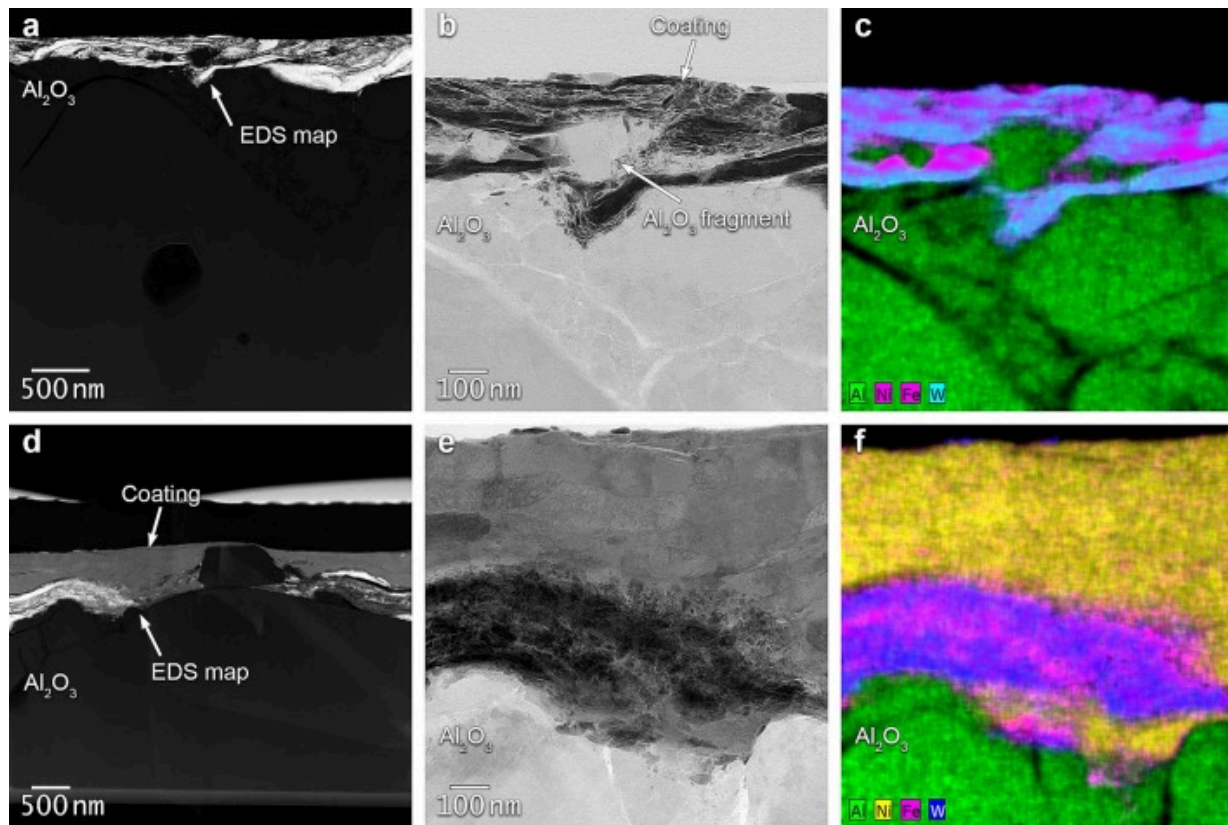


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Fig. 3. Structure of the W-Al-(Ni,Fe) coating observed via TEM: (a) BF STEM image showing the typical composite structure at low magnification; (b) and (c) BF and HAADF STEM images providing a magnified view of the coating; (d) corresponding STEM-EDS elemental map illustrating the chemical composition and multicomponent nature of the coating.

For the fabrication of W-Ni coatings, a combination of spherical shots and cubic billets was employed for the first time, demonstrating the effectiveness of cubic billets as a source of coating material. This approach expands the range of applicable materials for deposition, as cubic billets can be easily fabricated from various materials, providing a cost-effective alternative to spherical shots. In the specific case of W-Ni coatings, the use of cubic billets significantly enhanced the Ni fraction compared to spherical shots. As shown in our previous experiments [36,37], Ni welding to the substrate from spherical shots was limited, resulting in low Ni content. In contrast, the sharp edges of the cubic billets, combined with the angular features of the ceramic substrate, promoted material transfer and adhesion, substantially increasing the Ni fraction in the coatings. The composition of the coatings was directly influenced by the ratio of milling media ([Table 1](#)). When W dominated the milling media, the resulting coatings contained up to 90wt% W with minor amounts of Ni. Conversely, equal proportions of W and Ni media produced coatings with a balanced distribution of tungsten and nickel, referred to as W-10Ni and W-50Ni, respectively. The W-Ni coatings exhibit a composite structure ([Fig. 4](#)), consisting of a mixture of crystalline phases corresponding to bcc W and fcc Ni ([Fig. 1f](#)). The variation in chemical composition is reflected in the relative intensity of the XRD peaks, indicating changes in the phase ratio of the components. Additionally, the relative intensity of the substrate and coating peaks in the [XRD patterns](#) suggests that the W-Ni coatings are slightly thinner than the W-Al-(Ni,Fe) coating ([Fig. 1f](#)), which is consistent with the coating thickness trends ([Table 1](#)).



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Fig. 4. Structure of the W-Ni coatings observed via TEM: (a) HAADF STEM image showing the structure of the W-10Ni coating; (b) BF STEM image highlighting an Al_2O_3 fragment embedded within the W-10Ni coating; (c) corresponding STEM-EDS elemental map illustrating the chemical composition; (d) HAADF STEM image of the W-50Ni coating structure; (e) BF STEM image providing a magnified view of the W-50Ni coating; (f) corresponding STEM-EDS elemental map showing the chemical composition.

The experimental results indicate that the volume fractions of coating components for the materials used in this study are primarily governed by the ratio of milling bodies used (balls and cubes), rather than by their total mass, as evidenced by the close correlation

between the number ratios of milling bodies and measured volume fractions ([Table 1](#)). The resulting phase composition is most likely to reflect a cumulative effect of repeated impact events, which scale statistically with the number of bodies of each type. Parameters such as impact frequency, the spatial distribution of milling bodies in the working space, and their contact probability with the substrate collectively contribute to the preferential incorporation of each material.

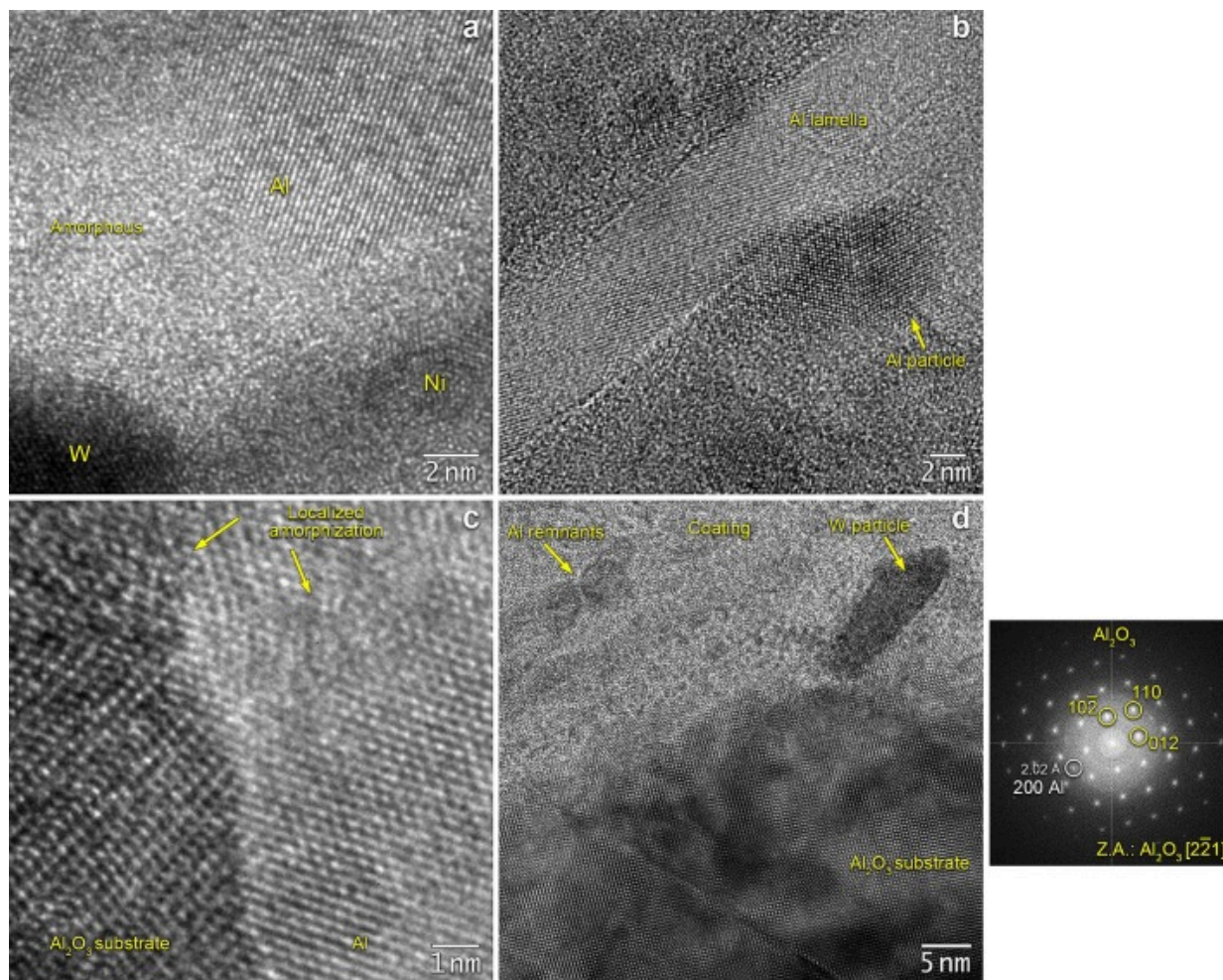
Atomic-scale structural integration

Having established the composite nature of the coatings and the incorporation of components during the coating process, it is essential to examine the distribution, intermixing, and interconnection of various components at the atomic scale. Such analysis provides key insights into the mechanisms driving the functional properties of the as-fabricated coatings.

In the W-Al-(Ni,Fe) coating, aluminum serves as a binder matrix, unifying the various components into a cohesive system ([Fig. 3a](#)). The individual counterparts (W, Ni, Fe, Cu) are present as flattened grains with varying degrees of deformation, likely governed by the mechanical characteristics of each component. W grains are larger due to their lower plasticity, while the more ductile Fe and Ni layers are thinner and exhibit a wavy morphology resembling fine veins, which are distinctly visible in the STEM images and the EDS map ([Fig. 3b, c, and d](#)). Additionally, fragments of the ceramic substrate can become embedded within the metallic matrix. During processing, larger grains of the substrate may locally fracture into smaller pieces or become dislodged as whole fragments ([Fig. 4a-c](#)), likely due to the movement of surrounding plasticized material along ceramic grain boundaries or cracks. These fragments are further pressed into and incorporated within the coating, directly enhancing its mechanical integrity.

The interconnection of the metallic counterparts within the coating occurs on the atomic scale. The presence of an amorphous structure at the interfaces ([Fig. 5a](#)), along with their diffuse nature, is a key indicator of highly efficient intermixing, interconnection, and

atomic-level interaction between the coating components [46]. The amorphous interfacial layer acts as a binding medium, providing a gradual transition across property mismatches and a complex mixture of chemical bonding [47]. This amorphous phase is a result of deformation-induced atomic intermixing of components under collisions [48,49] and is often developed during intense deformation in systems containing aluminum [37,38,50]. Overall, the matrix in the W-Al-(Ni,Fe) coating consists of a mixture of an amorphous phase and a crystalline structure with a strongly distorted lattice, interspersed with discrete aluminum particles, many of which are less than 10 nm in size (Fig. 5b), or aluminum lamellae thinner than 10 nm. Metallic particles of other components, such as W, are also observed within the matrix (Fig. 3b and c). Some of these particles are elongated, indicating that they form through the fragmentation of laminated structures.



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Fig. 5. HRTEM analysis of the W-Al-(Ni,Fe) coating: (a) amorphous phase formed between coating components; (b) matrix structure highlighting an Al particle and an Al lamella; (c) crystalline Al_2O_3 -Al interface showing localized amorphization; (d) amorphous Al_2O_3 -coating interface with the corresponding FFT pattern shown in the inset.

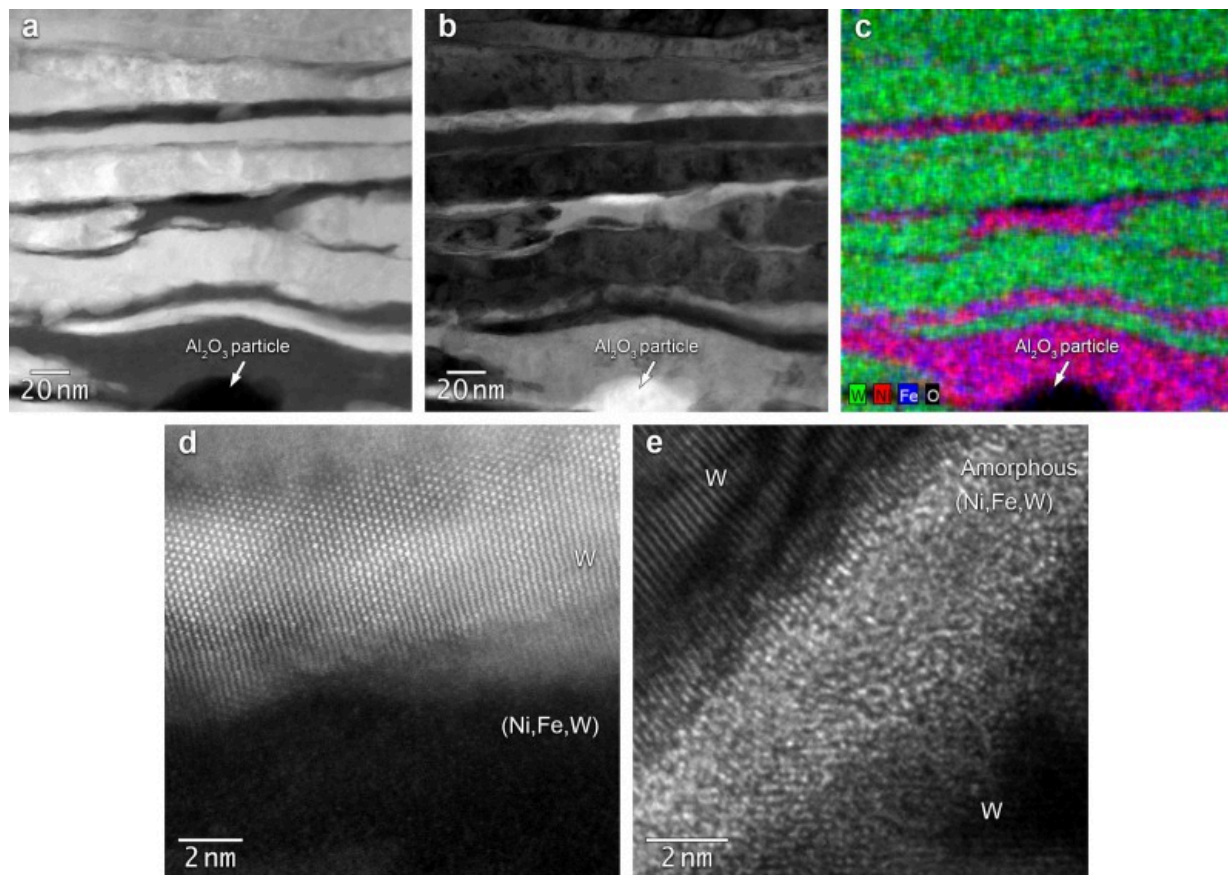
While the atomic-scale structure of the matrix phase provides valuable insights into the integration of the coating components, it is equally important to consider the structural characteristics of the interface between the coating and the alumina substrate. TEM analysis highlights the complex nature of this interface, revealing two distinct types of interfacial boundaries: crystalline (Fig. 5c) and amorphous (Fig. 5d). The crystalline structure at the ceramic–metal interface exhibits a continuous, dense transition between the atomic rows of Al_2O_3 and aluminum, indicating the formation of atomic-scale bonding. Meanwhile, localized amorphization is observed on both sides of the interface region (Fig. 5c). Disordered atomic clusters at the interface may compensate for lattice mismatches, reduce mechanical stresses, and thereby promote the formation of chemical bonds at the atomic scale [51].

By contrast, the amorphous boundary demonstrates a disordered structure, characterized by a diffuse and wavy interface (Fig. 5d). The crystalline structure of the Al_2O_3 phase at the interface is heavily strained, with locally distorted atomic rows, while the amorphous phase contains remnants of crystallinity. Measurements of interplanar spacings in these crystalline remnants are consistent with the (200) planes of the Al fcc phase. A distinct W particle welded to the interface is also observed, exhibiting a strongly distorted lattice. The fast Fourier transform (FFT) pattern shows a broad amorphous halo (Fig. 5d inset), indicating a non-uniform composition of the amorphous phase and the simultaneous amorphization of different components. Analysis of the amorphous halo reveals a lower boundary at $d \approx 3.5 \text{ \AA}$, corresponding to the interplanar spacing of the (120) planes of Al_2O_3 [52]. This suggests local amorphization of the ceramic substrate due to mechanical deformation. Local amorphization of ceramic materials under shock-induced or indentation-induced deformation has been widely reported [[53], [54], [55]]. The upper boundary of the halo, at $d \approx 2.02 \text{ \AA}$, corresponds to the (200) planes of Al, marking the lower limit of the amorphous region, which may originate from amorphous aluminum, tungsten, or other metallic components [52]. Although the amorphization of Al, potentially involving oxygen incorporation, may lead to a less dense structure and an increase in interplanar spacing, the amorphous phase at the interface is likely a mixture

of amorphized ceramic and aluminum. Such local amorphization of both materials may reduce barriers to mixing, facilitate atomic-scale chemical interactions, and play a crucial role in the bonding of dissimilar components.

In addition, some structural features of W-Ni coatings should be highlighted. In the W-10Ni coating, W serves as the primary structural component, while the more ductile Ni acts as a binder phase (Fig. 4a and b). This coating is characterized by coarse flattened W grains, which are bound together by a binding medium primarily composed of a mixture of Ni and Fe, along with small fragments of the W phase (Fig. 4b and c). Unlike the W-10Ni coating, the W-50Ni coating contains nearly equal amounts of Ni and W phases (Table 1). The structure can be described as randomly alternating layers of pure W and Ni with coarse flattened grains (Fig. 4d and e), interspersed with regions containing a mixture of these components alloyed with Fe as an additional constituent (Fig. 4f). This arrangement indicates localized random welding of Ni and W fragments, followed by their incorporation into a uniform composite structure and their potential intermixing at the interphase boundaries (Fig. 4e and f). As illustrated in Fig. 4f, the iron component is introduced alongside the harder W phase, likely due to abrasion of Fe from the sleeves during processing. A nanolaminated composite structure (Fig. 6a and b) and intermixing develop throughout the micro-forging process as different materials co-deform and penetrate each other along grain boundaries under the influence of shearing forces, as detailed in Ref. [56]. Overall, Ni and Fe exhibit rapid deformation-induced interdiffusion, resulting in the formation of a wide range of solid Ni-Fe solutions [56,57]. In the W-Ni coatings, EDS analysis revealed that the binding interface region contains, on average, 60–65wt% Ni, 15–20wt% Fe, and 15–20wt% W (Fig. 6c), which aligns with the equilibrium solubility of W in Ni, reaching up to 28wt% at room temperature [58]. The contrast observed in the HAADF STEM image (Fig. 6d), which reflects chemical composition [59], indicates compositional variations in the atomic rows of the W phase, suggesting that deformation-induced interdiffusion and atomic interactions also involve the W component. The amorphous phase in the W-Ni coatings was less pronounced compared to the W-Al-(Ni,Fe) coating and was mainly detected in interphase regions as thin

interlayers (Fig. 6e), composed of a mixture of Ni, Fe, and W. This suggests that the Al component appears to have played a more significant role in the amorphization process.



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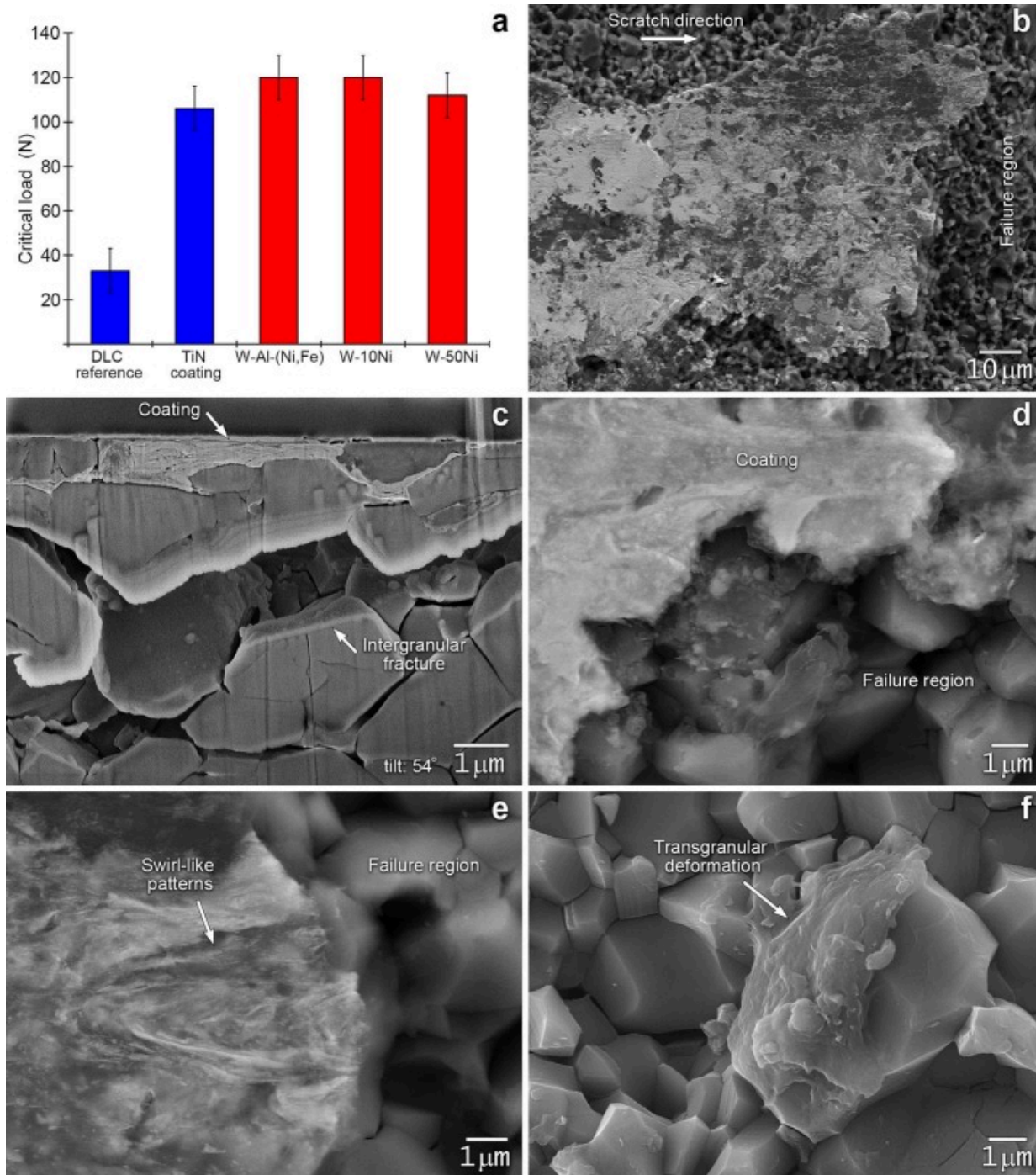
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Fig. 6. Structural features of the W-50Ni coating: (a) and (b) HAADF and BF STEM images showing the W-layered structure with Ni-Fe interlayer phases; (c) corresponding STEM-EDS elemental map illustrating the chemical composition and intermixing; (d) HAADF STEM image revealing compositional variations in W atomic rows; (e) HRTEM image illustrating the formation of an amorphous interlayer.

Properties of the as-fabricated coating

Scratch test for adhesion quality

[Fig. 7a](#) presents scratch test data for various materials, including reference [DLC](#) and TiN coatings, where the critical load is defined as the point at which coating [delamination](#) initiates. However, no delamination of the as-fabricated metallic coatings on ceramics was observed, even at loads of 100–120N, which correspond to the onset of ceramic substrate failure ([Fig. 7b](#)), indicating the strong adhesion between the coating and the substrate [47]. This strong adhesion is further evidenced in [Fig. 7c](#), which illustrates a cross-sectional image at the point of ceramic failure, showing [intergranular fracture](#) at depth, while the metallic coating remains adhered to the surface. The image highlights the complex mechanical interlocking between the coating and the substrate, which likely contributes to the exceptional adhesion of the [metallic films](#). All the as-fabricated coatings, including W-Al and W-Ni systems, demonstrate similar behavior and exhibit comparable adhesion characteristics during scratch deformation ([Fig. 7](#)).



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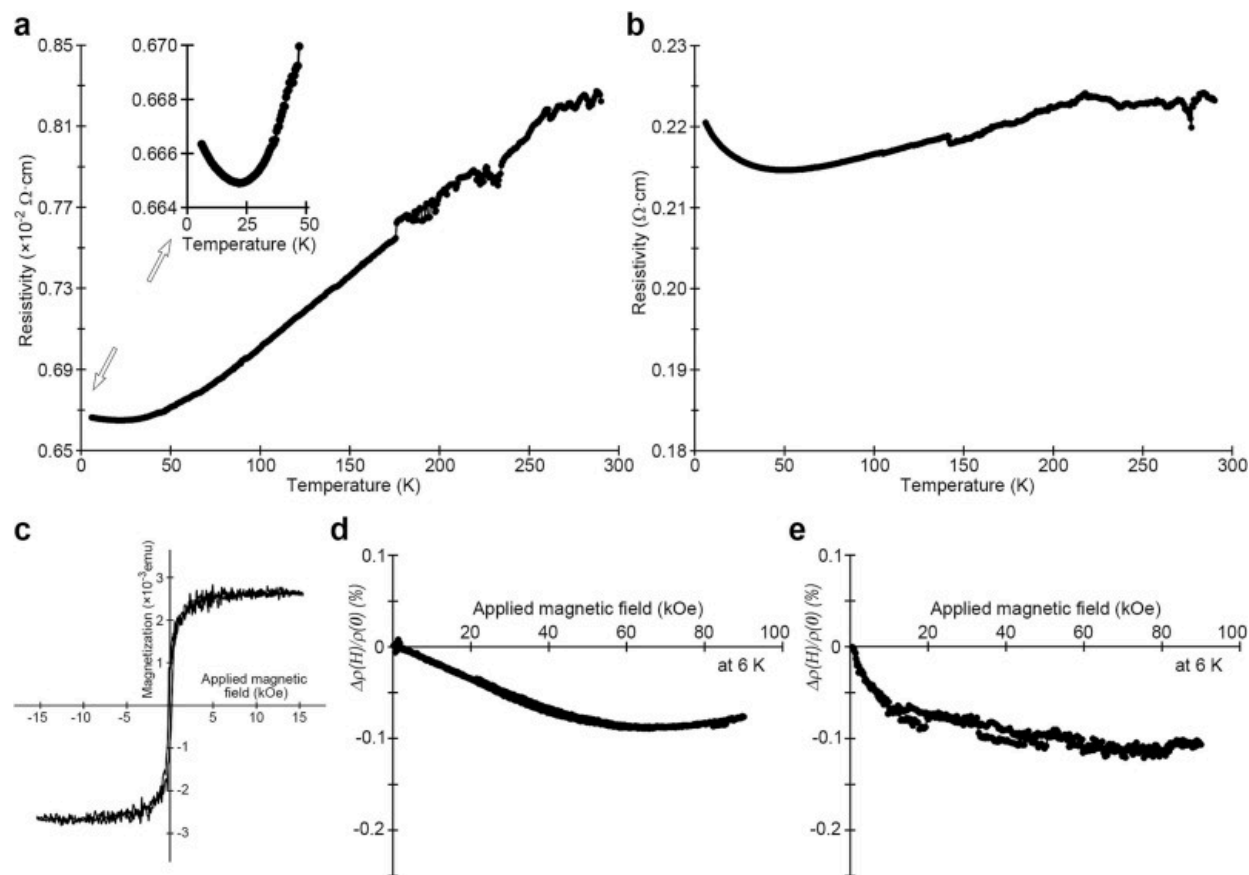
Fig. 7. Scratch test analysis of the as-fabricated coatings: (a) critical load values for reference DLC and TiN coatings, indicating delamination, and for as-fabricated metallic coatings, corresponding to the onset of ceramic substrate failure; (b) SEM image showing failure of the W-Al-(Ni,Fe) sample at the critical point; (c) cross-sectional FIB-prepared SEM image of the failure region in the W-Al-(Ni,Fe) sample; (d) enlarged SEM view of the failure region in the W-50Ni sample; (e) swirl-like patterns at the interface region in the W-Al-(Ni,Fe) sample; (f) transgranular deformation and localized plastic flow within a ceramic grain.

Overall, during the scratch test, the as-fabricated metallic coating gradually undergoes wear and fractures in conjunction with ceramic particles, without delaminating (Fig. 7d and 7e). The SEM images, showing deformation and fracture zones (Fig. 7d and 7e), reveal that the coating and substrate respond in a synchronized manner, functioning as an integrated system under scratch deformation. Fracture mechanisms in ceramics vary depending on the grain size [60]. In fine-grained regions, where grain sizes typically range from 1 to 3 μm , failure predominantly occurs through intergranular fracture (Fig. 7c). In contrast, larger ceramic grains exhibit transgranular deformation prior to failure, indicating localized plastic flow within individual grains during deformation (Fig. 7f). It is well-established that brittle materials can undergo permanent deformation under static contact if the applied load remains below critical values [61]. Our SEM observations reveal evidence of localized plastic deformation in the ceramic substrate during scratch deformation at loads below the failure threshold (100–120N). This deformation manifests as smoothing of the ceramic surface layers and the formation of swirl-like patterns at the interface regions (Fig. 7b and e). This smoothing can be attributed to tribologically-induced plastic flow, predominantly localized within thin surface layers [61], where synchronized deformation of the substrate and coating promotes intermixing and leads to the development of swirl-like patterns (Fig. 7e).

The co-deformation of the metallic coating and ceramic substrate, without delamination, demonstrates not only strong adhesion but also the high mechanical durability of the coating, capable of withstanding loads that lead to ceramic failure. Furthermore, under scratch deformation, the coating becomes further integrated into the microstructure of the substrate through synchronized plastic flow and mechanical interlocking. This behavior underscores the exceptional structural and mechanical integrity of the as-fabricated coating-substrate systems.

Functional properties

The functional properties of the as-fabricated coatings were investigated through electrical resistivity, magnetization, and magnetoresistance measurements to assess their potential applicability. Notably, the fabrication of metallic coatings transformed the initially insulating ceramic substrate into a conductive material, enabling electrical transport measurements (Fig. 8a and b). The temperature dependence of resistivity exhibits a metallic trend, as resistivity increases with temperature (Fig. 8a and b). Some fluctuations above 150K can be attributed to structural defects, amorphous phases, and local heterogeneities, which are characteristic of the as-fabricated multicomponent systems (Fig. 3, Fig. 4, Fig. 5, Fig. 6). However, in the temperature range of 6–50K, a distinct upturn in resistivity is observed in the W-50Ni coating (Fig. 8b), consistent with an increase in resistivity as temperature decreases. This effect is less pronounced in the W-Al-(Ni,Fe) coating but can still be discerned in the range of 6–25 K upon detailed analysis (Fig. 8a inset). The observed variation in resistivity behavior between W-Al-(Ni,Fe) and W-Ni coatings can be attributed to their distinct chemical and phase compositional differences along with structural variables. Despite being relatively higher than that of pure metals, the resistivity remains sufficient to form a conductive layer on the ceramic substrate. This conductive layer can effectively serve as a grounding foundation for subsequent functional coatings, demonstrating the practicality of the developed technique.



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Fig. 8. Functional properties of the as-fabricated coatings: (a) temperature dependence of resistivity for the W-Al-(Ni,Fe) coating, with an enlarged view of the low-temperature region shown in the inset; (b) temperature dependence of resistivity for the W-50Ni coating; (c) magnetization curve of the W-50Ni coating; (d), (e) magnetoresistance behavior of the W-Al-(Ni,Fe) and W-50Ni coatings, respectively.

The initially non-magnetic ceramic substrate becomes magnetically active due to the incorporation of ferromagnetic phases within the coatings (Fig. 3, Fig. 4, Fig. 6). This effect is most pronounced in the W-Ni coatings due to the larger fraction of the ferromagnetic

Ni phase in the system ([Table 1](#)), which enhances magnetic interactions. The narrow hysteresis loop suggests a low coercive force and low remanent magnetization, which are characteristic features of soft magnetic materials ([Fig. 8c](#)). These characteristics reflect the material's ability to be easily magnetized and demagnetized under relatively weak external magnetic fields. Although the magnetic response differs from that of bulk coarse-grained ferromagnetic materials, the combination of soft magnetic behavior, high thermal conductivity of tungsten, and strong adhesion to ceramics positions these coatings as promising candidates for multifunctional material design.

Magnetoresistance refers to the change in electrical resistivity of a material under an applied magnetic field. In the as-fabricated coatings, negative magnetoresistance was observed at 6K, indicating a reduction in electrical resistivity with increasing magnetic field ([Fig. 8d and e](#)). Both W-Al-(Ni,Fe) and W-50Ni coatings exhibit a similar trend, demonstrating negative magnetoresistance over the entire measured range up to 90 kOe. However, in the W-Al-(Ni,Fe) sample, the magnetoresistance behavior is more stable across the magnetic field range and decreases more gradually with increasing field, whereas the W-50Ni sample shows greater fluctuations and a more rapid drop in magnetoresistance ([Fig. 8d and e](#)). This magnetoresistance phenomenon is likely due to the presence of magnetic Fe and Ni phases within the coatings. Notably, pure Fe and Ni in a coarse-grained bulk state or in solid solutions are not expected to exhibit negative magnetoresistance under standard conditions, indicating that the structural configuration of the material plays a key role. Overall, these effects indicate that the observed magnetoresistance behavior is governed by electronic interactions, likely involving quantum transport phenomena, warranting further discussion in the next section.

Discussion

To further understand the exceptional adhesion and functional properties of the as-fabricated coatings, it is essential to analyze the underlying mechanisms governing interfacial bonding and electronic interactions.

Interfacial bonding

Interfacial processes are driven by a combination of interrelated physical and chemical mechanisms. The deformation behavior of tungsten within the coatings provides indirect evidence of the localized temperature rise during processing. Despite its inherent brittleness, the W component exhibited ductile behavior, forming lamellar structures with thicknesses below 10nm ([Fig. 3](#), [Fig. 6](#)). Given that the ductile-to-brittle transition temperature of tungsten lies between 200–300°C [[62](#)], this suggests that the local processing temperature exceeded this threshold. Otherwise, W fragments would have fractured rather than flattened. Molecular dynamics simulations indicate that local impact temperatures at the point of contact within atomic clusters may reach extremely high values, depending on impact conditions [[63](#),[64](#)]. The low thermal conductivity of the ceramic substrate likely contributes to heat retention within the metallic fragments, further enhancing their plasticity and flowability during processing. The pressure at the contact point during impact can also reach extremely high values, leading to significant local stress concentrations [[64](#)]. The ability of metallic materials to densely fill substrate irregularities and penetrate along ceramic grain boundaries provides additional evidence of significant material plasticization and high-pressure conditions under impact ([Fig. 7c](#)). When plasticized metallic material penetrates intergranular grooves ([Fig. 2](#)), it experiences confined flow conditions, generating localized pressure within the intergranular space, which is even capable of displacing original ceramic grains into the forming coating ([Fig. 4b](#)). Such high-pressure conditions, along with elevated temperatures, facilitate intense plastic deformation, promote material flow, and enhance interfacial atomic interactions.

In most cases of metal/ceramic interface interactions, oxygen atoms play the primary reactive role, forming covalent or ionic bonds with metal atoms in the coating [[65](#)], [[66](#)], [[67](#)]]. In the polycrystalline structure of the Al₂O₃ substrate, grains are randomly oriented, exposing various crystallographic facets at the surface ([Fig. 2](#)). Since the majority of these facets are oxygen-terminated, numerous reactive sites are present, allowing chemical interactions between oxygen atoms from the ceramic and metal atoms in the coating. The

oxygen partial pressure during processing can further enhance interfacial reactions by increasing oxygen activity at the substrate surface [65,67]. The atmospheric processing environment, combined with the localized increase in temperature and pressure induced by impact, promotes interfacial chemical reactions by facilitating oxygen bonding with metal atoms in the coating and further strengthening adhesion, which is inherently governed by the nature of atomic bonds [47].

It is also important to consider that in the polycrystalline structure of the ceramic substrate, various crystallographic facets may be exposed at the surface, some of which can be aluminum-terminated. This suggests not only metal–oxygen bonding but also metal–metal bonding at localized interfacial regions [66,67], or even mixed metallic-ionic bonding, where the ionic component arises due to the electronegativity differences between Al and other metallic components. Localized deformation-induced amorphization at the ceramic–metal interface (Fig. 5) and atomic intermixing within the interfacial amorphous region may further promote chemical interactions between atoms, resulting in a mixed bonding character. Additionally, such amorphous interfacial layers accommodate strain more effectively than crystalline interfaces, thereby mitigating stress concentrations and enhancing structural stability [51]. Overall, the combination of complex chemical bonding and mechanical interlocking at the interface accounts for the efficient coating formation and the high adhesion strength observed in the as-fabricated coatings (Fig. 7).

Structural features and electronic interactions

In the temperature range of 6–50K, an upturn in resistivity is observed, with a more pronounced effect in the W-50Ni coating, which has a higher nickel content (Fig. 8, Table 1). Additionally, both samples exhibit negative magnetoresistance, although their response to the applied magnetic field follows different trends (Fig. 8). The W-Al-(Ni,Fe) coating demonstrates more stable magnetoresistance behavior with a gradual decrease in resistivity as the field increases, whereas the W-50Ni sample shows greater fluctuations and a more rapid drop in resistivity. The observed increase in resistivity at low

temperatures and the presence of negative magnetoresistance can be explained by the theory of weak localization, as both phenomena are interconnected, while variations in behavior are likely associated with differences in structural and phase configuration.

Weak localization is a quantum interference effect in disordered electronic systems, where multiple scattering events cause constructive interference of electron wavefunctions, leading to increased electrical resistivity at low temperatures [[68], [69], [70], [71], [72]]. This phenomenon is influenced by the complex and integrated multiphase structure of the as-fabricated coatings, which combines amorphous regions, nanocrystalline grains, and lamellar morphology into a unified system (Fig. 3, Fig. 4, Fig. 5, Fig. 6). Additionally, structural imperfections such as highly distorted lattices, remnants of crystallinity, the coexistence of multiple magnetic phases, non-equilibrium solid solutions, and a high density of point defects and impurity atoms further contribute to disorder (Fig. 5, Fig. 6). These structural characteristics align with findings in the literature describing weak localization in disordered scattering media [[68], [69], [70], [71], [72]]. The stronger weak localization effect in the W-50Ni coating manifests as a greater increase in resistivity at low temperatures (Fig. 8). This can be attributed to the high nickel content and the localized formation of Ni-Fe heterogeneous magnetic phases (Fig. 6), which create additional scattering centers, enhancing quantum transport phenomena and increasing resistivity fluctuations [[68], [69], [70], [71], [72]].

Negative magnetoresistance, characterized by a decrease in resistivity under the application of a magnetic field (Fig. 8), is consistent with the quantum-interference origin of the observed effects. The reduction in resistivity under a magnetic field is associated with the suppression of weak localization, as reported in numerous experimental and theoretical studies [[68], [69], [70], [71], [72]]. Notably, in the W-Al-(Ni,Fe) coating, negative magnetoresistance exhibits a more gradual, linear-like response to magnetic field, suggesting that quantum interference effects are more robust and require higher fields for suppression (Fig. 8d). In contrast, the W-50Ni sample shows a more pronounced, nonlinear (logarithmic-like) magnetoresistance behavior, indicating that weak localization is more easily disrupted by the applied magnetic field (Fig. 8e) [68,69].

The noisy behavior of the magnetoresistance in W-50Ni may be linked to the local spatial inhomogeneity and compositional fluctuations within the magnetic phases (Fig. 4, Fig. 6), which affect magnetic scattering [68,69]. The observed differences in magnetoresistance behavior can be attributed to underlying structural features. In the W-Al-(Ni,Fe) system, where structural disorder is more pronounced and the magnetic phase content is lower, weak localization appears more resilient under applied magnetic fields (Fig. 8d and e). This suggests that structural disorder may help preserve quantum interference effects under the influence of magnetic fields, which could be important for the design of materials for quantum-coherent applications.

In addition, the chemical composition of the coatings could similarly contribute to unique electronic properties. In amorphous Al-W systems, a negative temperature coefficient of resistivity has been reported [73], aligning with observations in similarly disordered metallic systems. This behavior is associated with enhanced electron scattering due to structural disorder, which suppresses conventional metallic conductivity and is closely tied to quantum interference effects, such as weak localization [68,69]. These structural characteristics may further promote the emergence of additional quantum phenomena, as evidenced by the observation of superconductivity in amorphous Al-W alloys at specific compositions ($T_c \approx 1.8$ K) [73]. Although superconductivity was not detected in the studied coatings, this example highlights the potential for exploring other quantum states in similarly disordered metallic systems with tailored compositions and structural features.

The emergence of quantum phenomena in multicomponent metallic systems with nanostructured phases warrants further investigation, as similar effects may arise universally in materials subjected to intense plastic deformation. Understanding these effects could provide deeper insight into the quantum behavior of materials under extreme conditions and open new possibilities for designing advanced functional materials. This study demonstrates that weak localization and negative magnetoresistance can emerge in multiphase metallic structures fabricated on ceramic substrates using a simple mechanical alloying approach. These effects are directly linked

to the unique structural complexity mechanically integrated into a uniform system from the atomic to the microscopic level. The observation of such quantum phenomena in structurally intricate metallic systems highlights the fundamental interplay between microstructure and electronic behavior, offering valuable experimental insight into how material configuration influences quantum transport. The ability to engineer complex metallic structures with controlled electronic and magnetic interactions opens new avenues for developing next-generation materials, particularly in spintronics, magnetic sensing, and other quantum-enabled technologies. This underscores the potential of tailored structural design as a powerful tool for creating novel quantum materials where quantum effects can be systematically controlled and optimized for technological applications.

Conclusions

1. This study demonstrates an efficient mechanical alloying approach using ultrasonically driven shot impact equipment for metallizing ceramic substrates. The utilization of metallic cubic billets as milling media expands the range of materials that can be employed as coating sources.
2. The as-fabricated coatings impart electrical conductivity and magnetic properties to the initially insulating ceramic substrate, while their exceptional adhesion ensures mechanical integrity, allowing the coating to co-deform with the substrate without delamination prior to ceramic fracture.
3. The developed approach enables the mechanical tailoring of multiphase composition and structure. The unique structural complexity, combining amorphous regions, nanocrystalline grains, and localized non-equilibrium solid solutions, is mechanically integrated into a uniform system from atomic to microscopic level during processing.

4. The as-fabricated coatings exhibit quantum transport phenomena in the temperature range of 6–50 K, including weak localization and negative magnetoresistance observed at 6 K, which are strongly correlated with their structural complexity. These findings highlight the interplay between microstructure and electronic behavior, paving the way for further exploration of quantum phenomena in disordered multicomponent metallic systems fabricated under intense plastic deformation.

CRediT authorship contribution statement

S. Romankov: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Y.C. Park:** Writing – review & editing, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition. **R.Y. Umetsu:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **S.V. Komarov:** Writing – review & editing, Visualization, Investigation. **D.V. Louzguine-Luzgin:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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